

CHE 555 – Homework 1

Problem 1

$$\begin{aligned}x^* &= -0.363804 \\ y(x^*) &= 0.173294 \\ D_{min} &= 0.402970\end{aligned}$$

Problem 2

Quantity	Value
Base length x^*	12.5992 cm
Height h^*	6.2996 cm
Surface area S^*	476.2203 cm ²

Problem 3

Degrees of freedom = 1

Problem 4

Case	Value
Min cost production	300 barrels/day
Cost per barrel	110.00
Max profit production	1250 barrels/day
Profit	147,250.00/day

Problem 5

Best model: exp-quadratic

CHE 555 HW1 - Problem 5: Curve Fitting

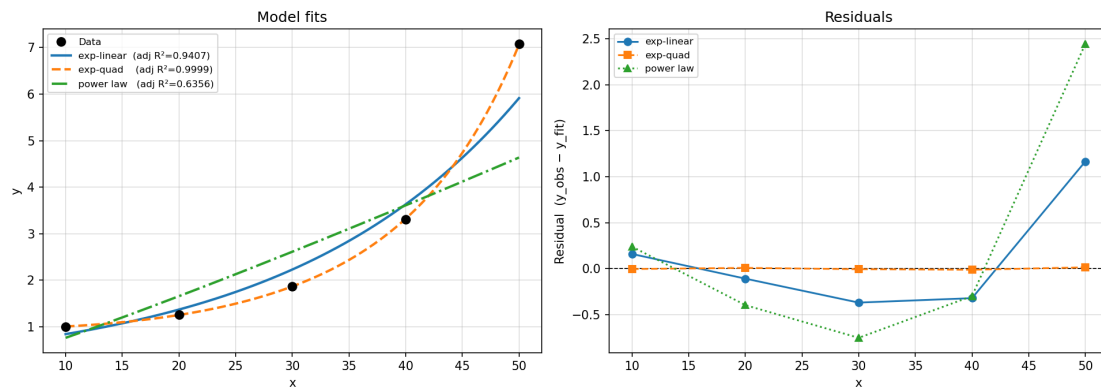


Figure 1: Curve fitting results